CS2201 Computer Organization & Architecture

L18: Introduction to Cache Memory



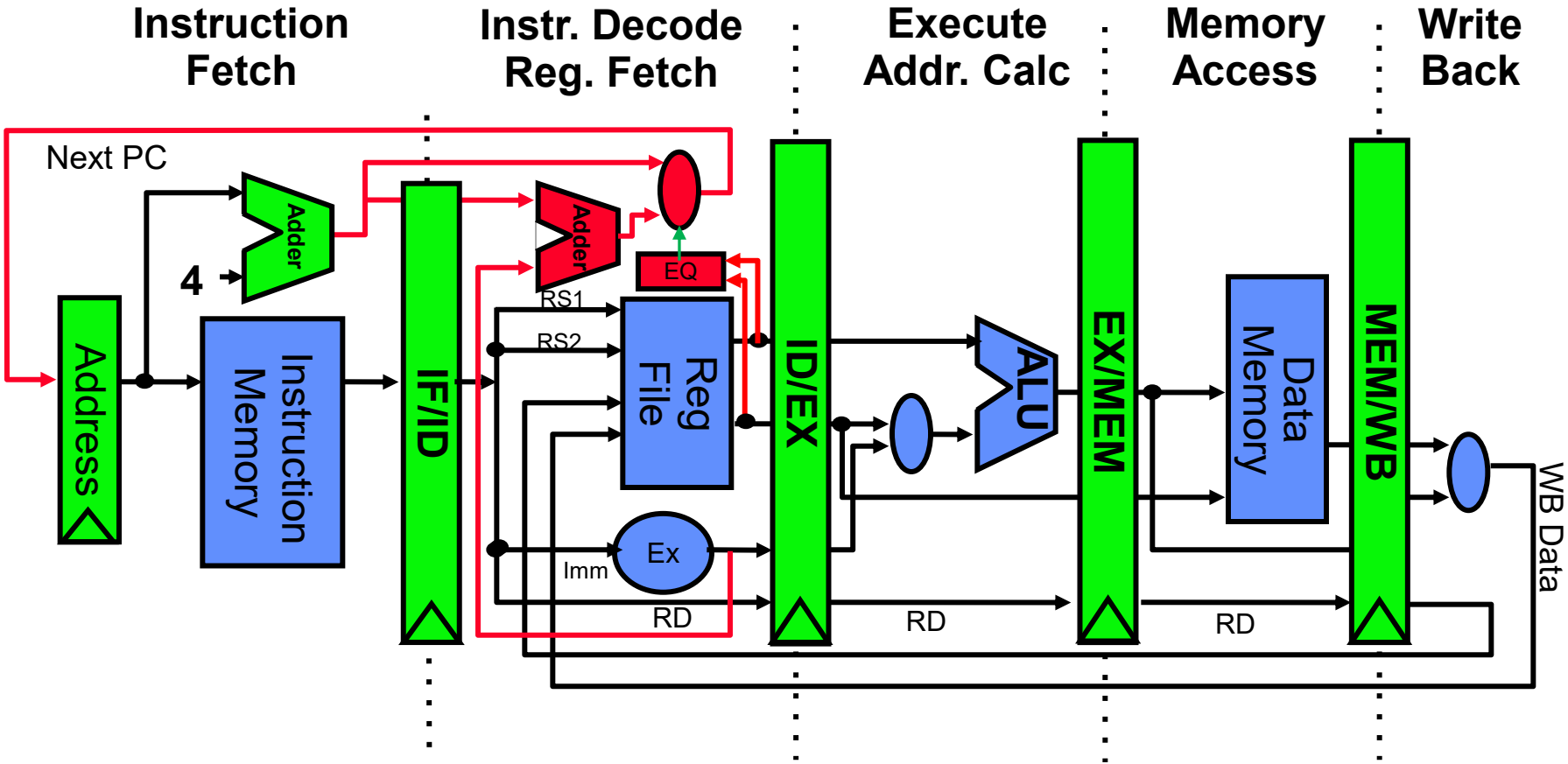
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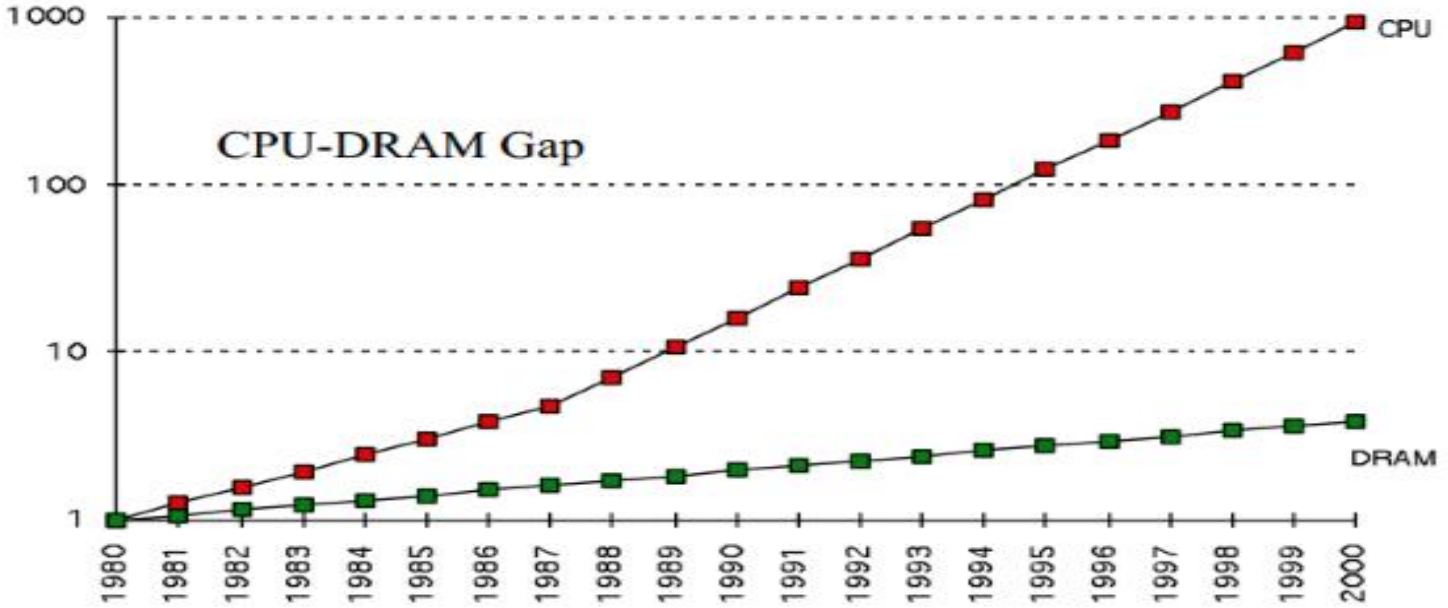
Indian Institute of Technology Guwahati

Pipelined RISC Data path



Processor Memory Performance Gap

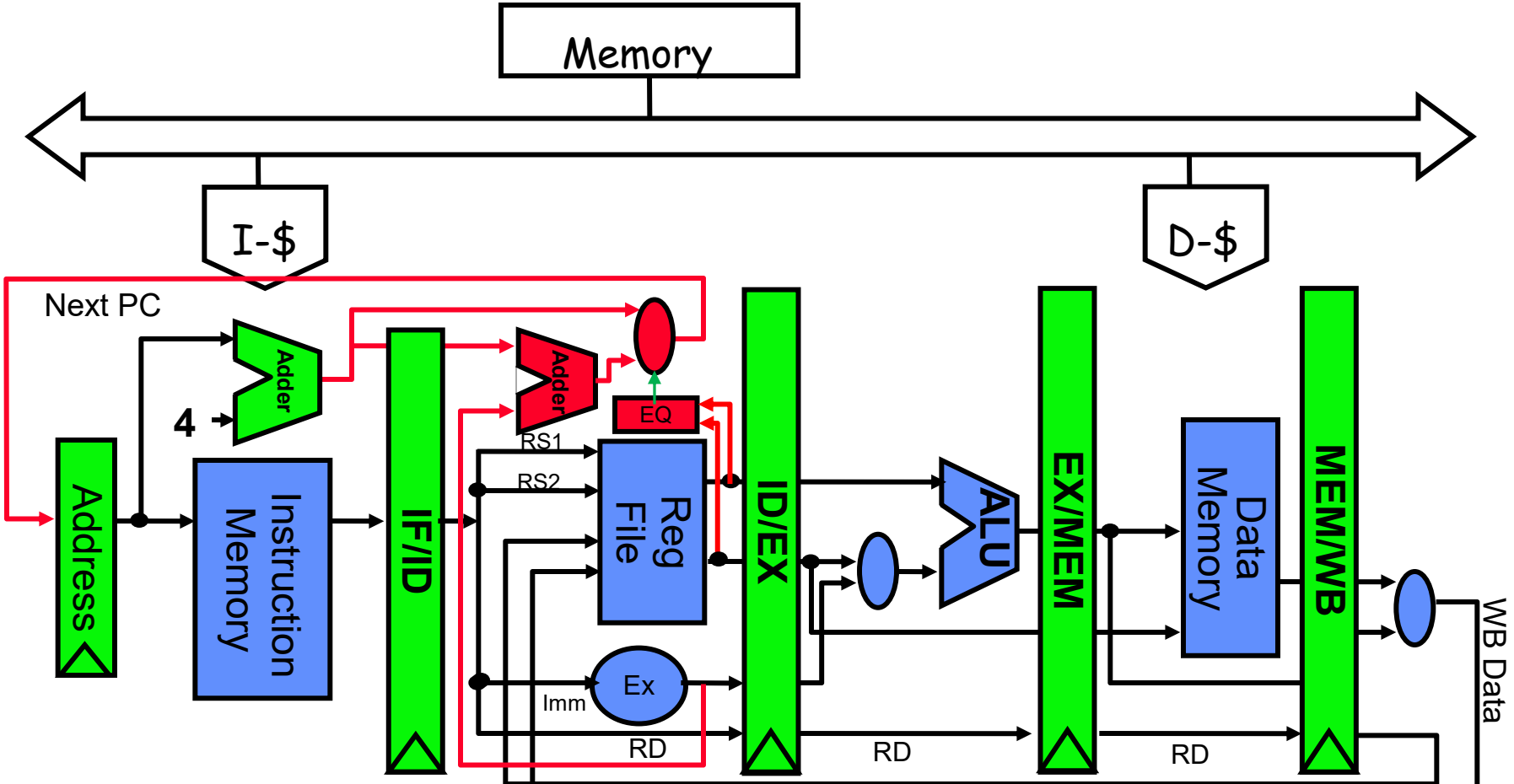
- Processor vs Memory Performance



1980: no cache in microprocessor;

1995 2-level cache

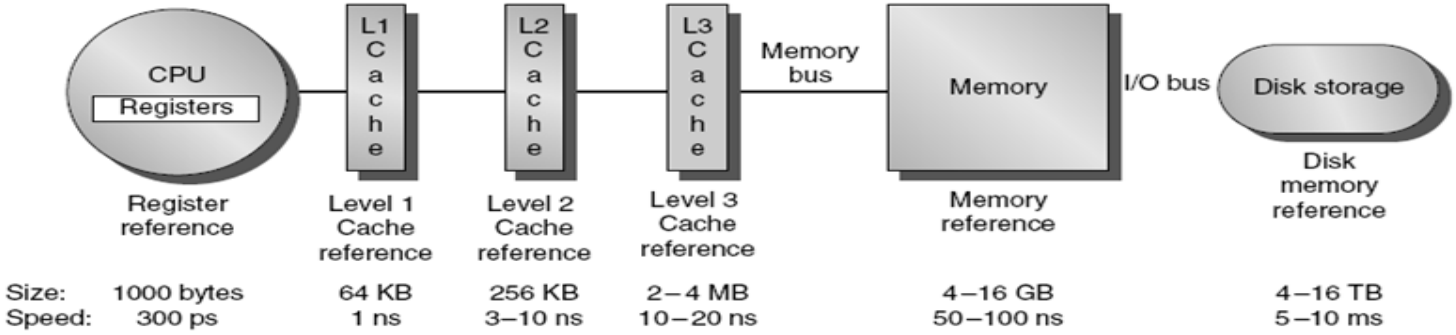
Relationship of Caches and Pipeline

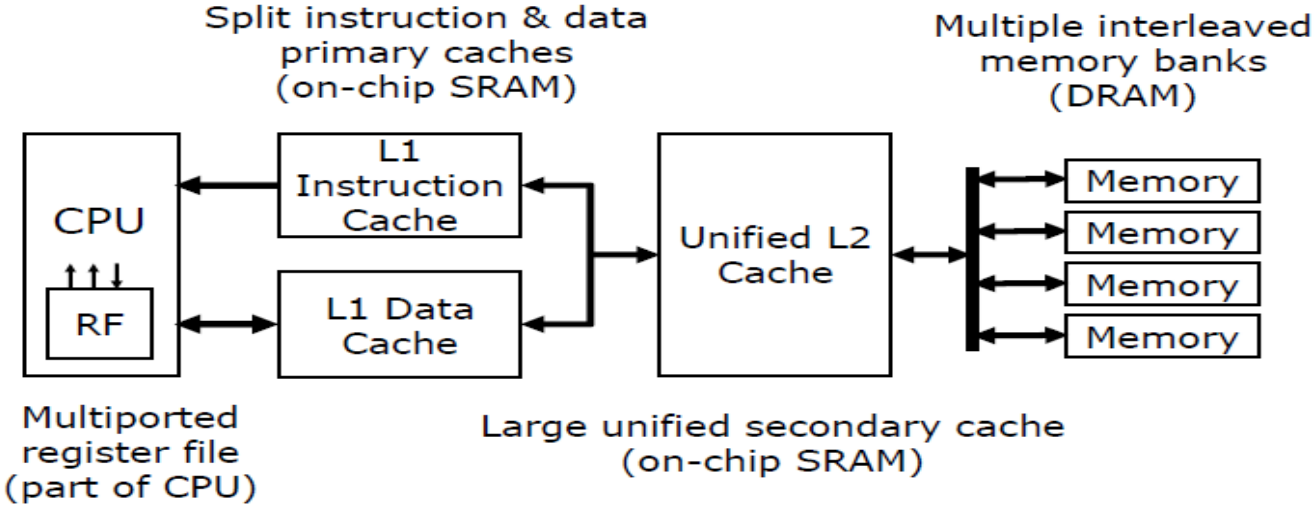


Role of memory

- Programmers want unlimited amount of fast memory
- Create the illusion of a very large and fast memory
- Implement the memory of a computer as a hierarchy
- Multiple levels of memory with different speeds and sizes
- Entire addressable memory space available in largest, slowest memory
- Keep the smaller and faster memories close to the processor and the slower, large memory below that.

Memory Hierarchy

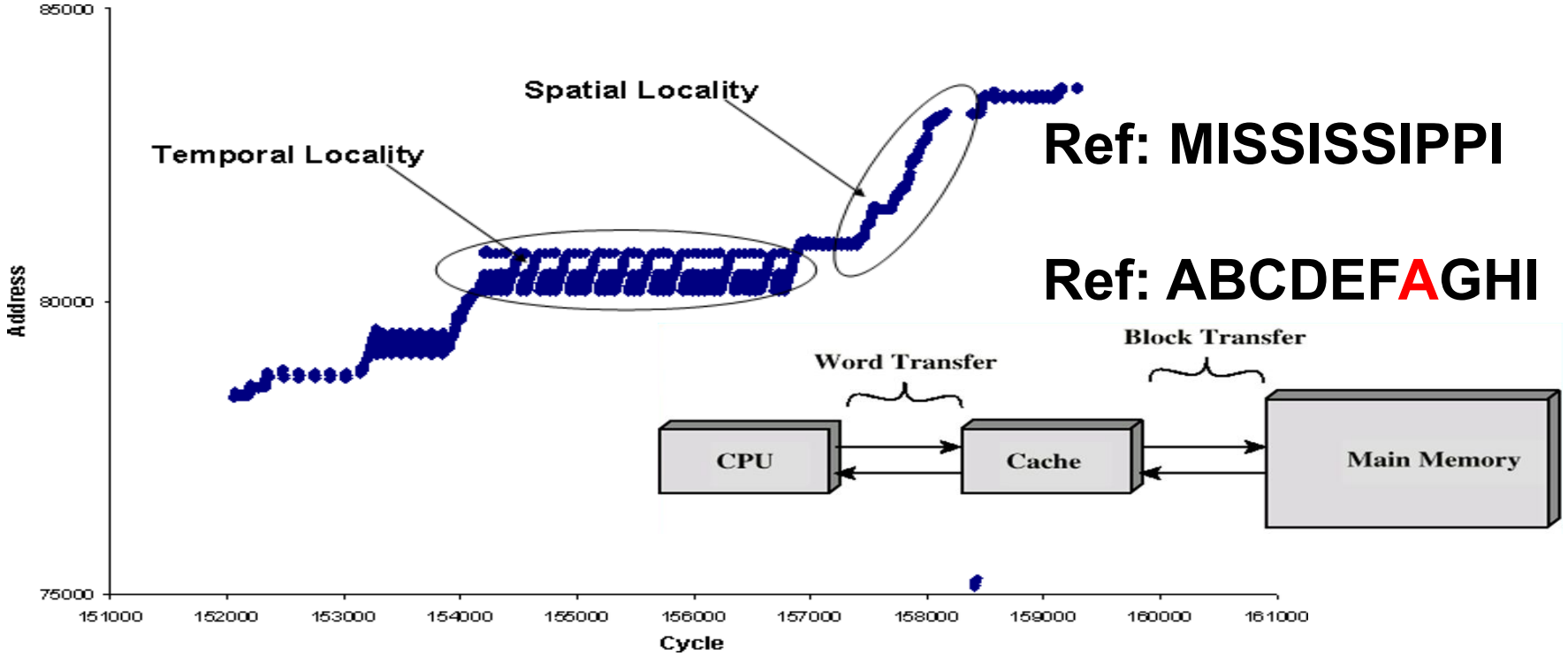




Cache Memory - Introduction

- Cache is a small, fast buffer between processor and memory
- Old values will be removed from cache to make space for new values
- **Principle of Locality :** Programs access a relatively small portion of their address space at any instant of time
- **Temporal Locality :** If an item is referenced, it will tend to be referenced again soon
- **Spatial Locality :** If an item is referenced, items whose addresses are close by will tend to be referenced soon

Access Patterns



Ref: MISSISSIPPI

Ref: ABCDEFAGHI

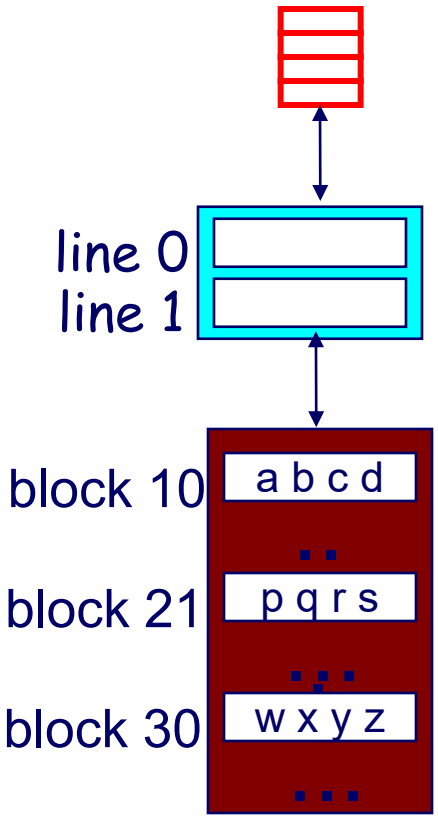
Cache Fundamentals

- **Block/Line :** Minimum unit of information that can be either present or not present in a cache level
- **Hit :** An access where the data requested by the processor is present in the cache
- **Miss :** An access where the data requested by the processor is not present in the cache
- **Hit Time :** Time to access the cache memory block and return the data to the processor.
- **Hit Rate / Miss Rate:** Fraction of memory access found (not found) in the cache
- **Miss Penalty :** Time to replace a block in the cache with the corresponding block from the next level.

CPU – Cache Interaction

❖The transfer unit between the CPU register file and the cache is a 4-byte word

❖The transfer unit between the cache and main memory is a 4-word block (16B)



❖The tiny, very fast CPU register file has room for four 4-byte words

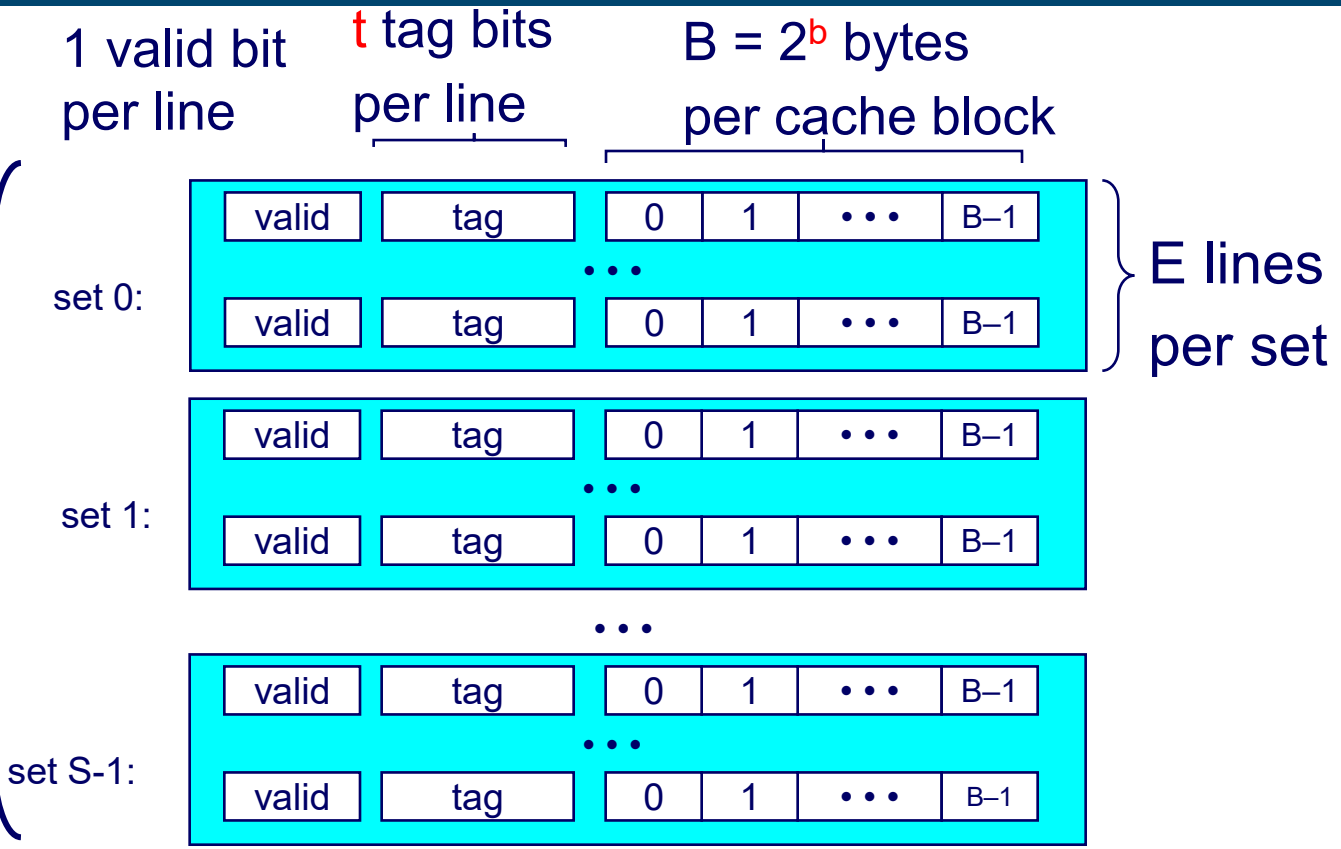
❖The small fast **L1 cache** has room for two 4-word blocks

❖The big slow main memory has room for many 4-word blocks

General Organization of a Cache

- Cache is an array of sets
- Each set contains one or more lines
- Each line holds a block of data

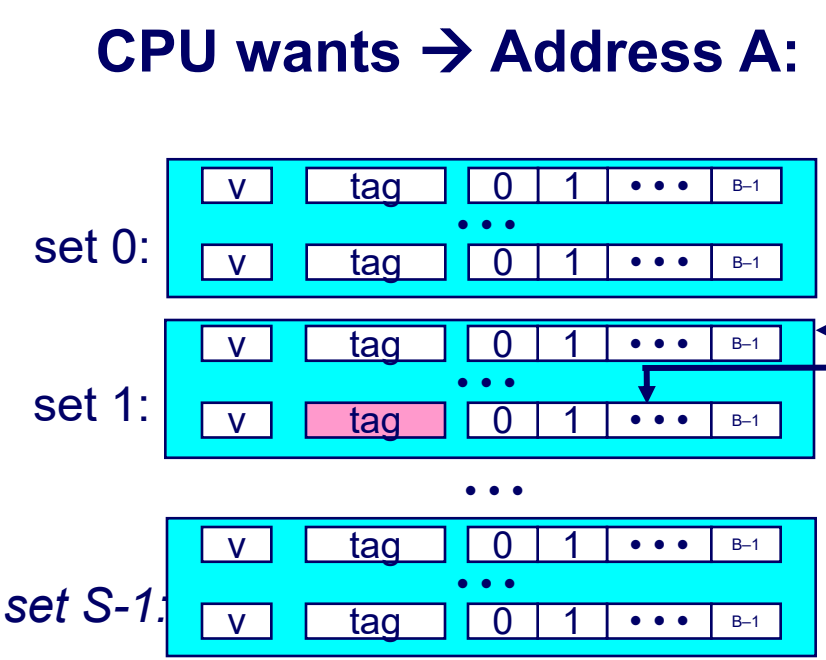
$S = 2^s$ sets

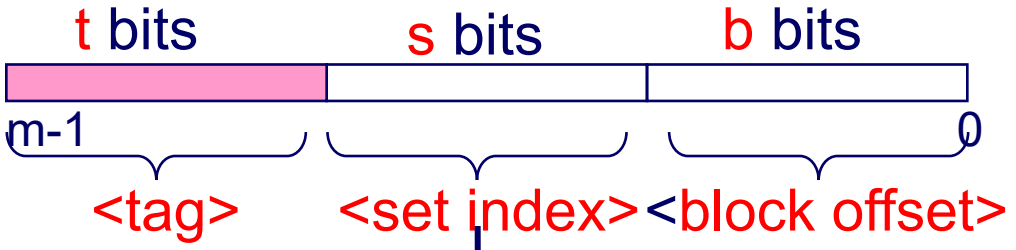


Cache size: C = B x E x S data bytes

Addressing Caches

CPU wants → Address A:



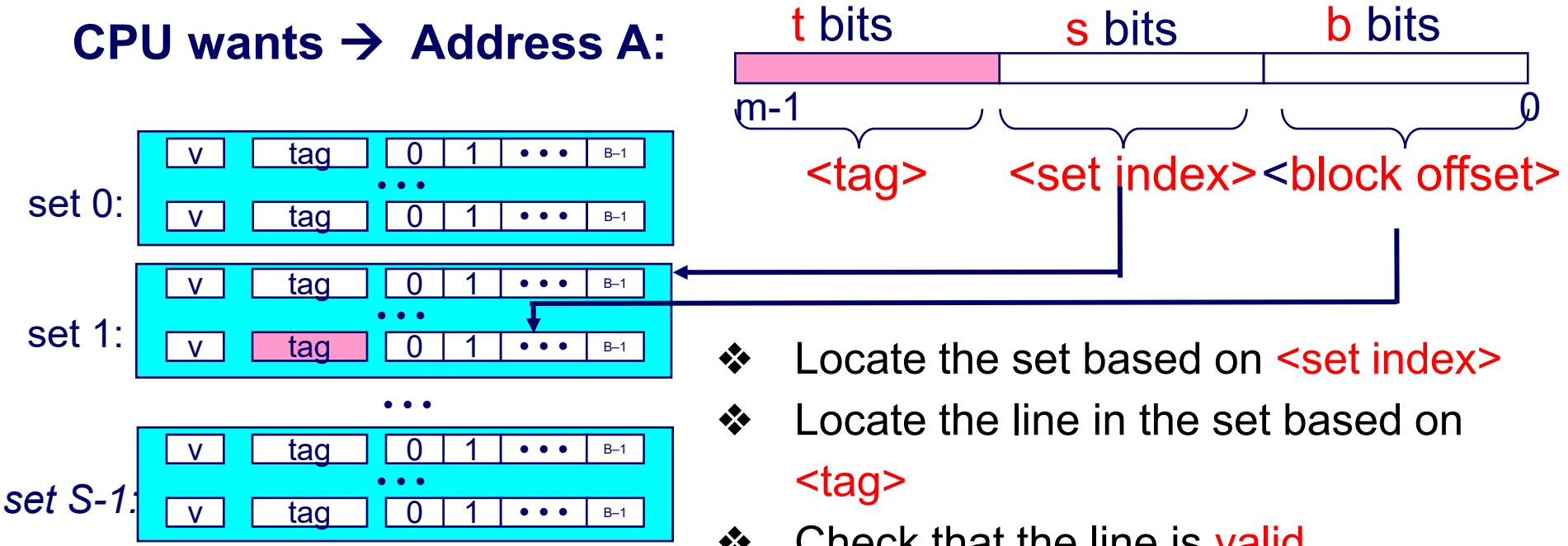


❖The word at **address A** is in the cache if the tag bits in one of the <valid> lines in set <set index> match <tag>

❖The word contents begin at offset <block offset> bytes from the beginning of the block

Addressing Caches

CPU wants → Address A:

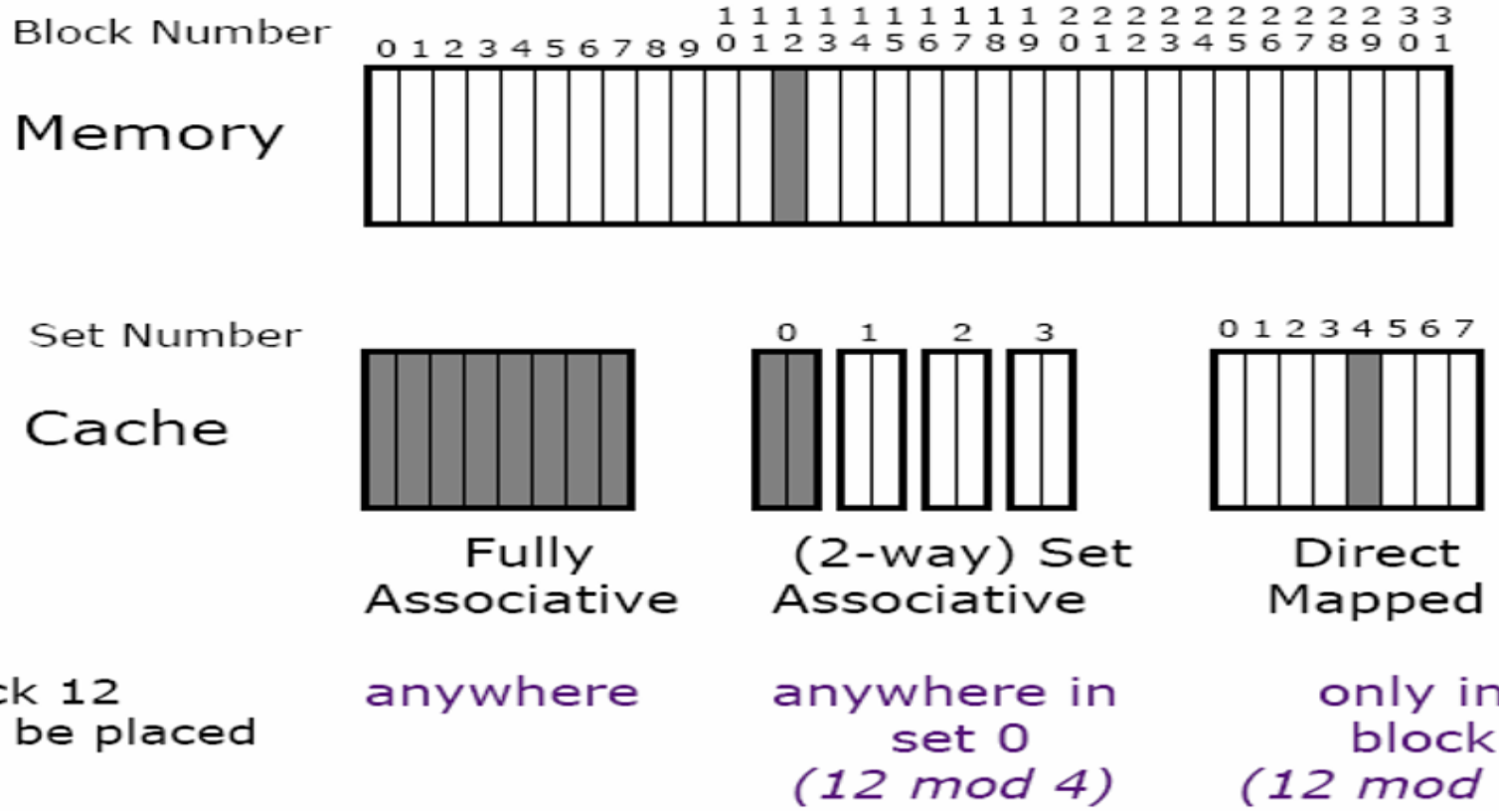


- Locate the set based on <set index>
- Locate the line in the set based on <tag>
- Check that the line is valid
- Locate the data in the line based on <block offset>

Four cache memory design choices

- Where can a block be placed in the cache?
  - **Block Placement**
- How is a block found if it is in the upper level?
  - **Block Identification**
- Which block should be replaced on a miss?
  - **Block Replacement**
- What happens on a write?
  - **Write Strategy**

Block Placement





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